

iRallyMeter

Audit and hardening report for Amirali · 5 August 2026
Prepared by Saam · branch SA-V3 at 90920ca

How to read this. Every number here is either something that was measured or is explicitly labelled as not measured. Several of them look similar and the difference matters, so the report keeps them apart rather than blending them into one confidence level.

1. Which branch to pull

```
git fetch origin
git checkout SA-V3
```

main is untouched at c151ca2 , exactly the commit you left. Nothing was force pushed and no history was rewritten. There is deliberately **no pull request open**, because opening and merging one is yours and Saam's call, not mine.

The work sits on two branches so you can review and take them separately.

Branch	What it holds	Size
SA-V2	Correctness fixes only. Nothing discretionary, nothing cosmetic.	123 files, +17,168 / -1,105
SA-V3	Branches off SA-V2 . Polish, test infrastructure, the first iOS build, and the fixes from two external code reviews.	50 files, +2,223 / -198

99 commits in total against main . Each one carries the reasoning and, where behaviour changed, the test that failed before the fix.

2. The metrics

	Your original version	Now
Unit and widget tests	354 passing	414 passing, 0 failing
Test files	7	43
Integration tests (real device)	none existed	5 of 5 green
Golden image tests	none existed	4, plus a determinism test
flutter analyze	2 issues	2 issues, the same two
iOS	never built	builds, installs, launches
Skipped tests	1	1, held at full strength

The two analyze issues are pre existing and unrelated to any of this: a deprecated `activeColor` , and an unused test parameter. The single skipped test was left at full strength rather than weakened to make the suite look clean.

One theme predicts most of what was found. The app asserted things it had not measured, and in most cases a documentation comment already described the correct behaviour while the code had quietly stopped implementing it. Your comments were usually right. The code had drifted away from them.

3. The defects that would have cost a crew a result

Ordered by how much damage each one does on a stage.

CRITICAL A single tap wiped your trip counter

Before

One tap anywhere on a trip tile zeroed it. That tile is roughly 40 percent of the cluster in landscape. The only guard was lock mode, which is off by default. One gloved brush on a gravel section and Trip A was gone mid stage.

After

Long press. The handler was renamed as well as rebound, because it was called `_confirmReset` and confirmed nothing, which is how it survived review.

Worth noting

Three things in the code disagreed with each other: the `onTap` binding, that method name, and the class documentation which already said *reset on long press*. The documentation was correct.

CRITICAL Zero distance through a tunnel, on some phones

Before

Receivers that report `0.0` for Doppler speed rather than null exist, and your own comment names them. Those phones measured open road correctly, then entered a tunnel anchored at zero speed and accrued nothing for its entire length, while the cluster showed a confident 0 km/h.

After

The emit site now applies the same validity test that already sat three lines above it.

Proof

Putting the old line back fails the new tunnel test and nothing else. This was verified by reverting it, not assumed.

CRITICAL The compass lied while you were stopped

Before

Once the car had moved, the GPS course stayed valid for the rest of the app's life. Stop, and the cluster showed a frozen old heading still labelled `GPS`, with nothing marking it stale. The magnetometer branch was unreachable.

After

The source releases below 0.8 m/s and reacquires above 1.4 m/s. The gap between those two numbers is deliberate. A single threshold would make the label flicker at the boundary, so the gap is the feature.

CRITICAL The first position came from the wrong provider

Before

The cold start seed called `getLastKnownPosition()` with no arguments, and that parameter defaults to false, so on any phone with Play Services it returned the fused provider. That is the road snapping provider the rest of the file deliberately refuses everywhere else.

Why it matters

A seed fix snapped onto a road is worse than a slightly stale raw one. It is confidently wrong, and it is the position the trip anchor starts from.

HIGH The moving average roughly doubled at 5 Hz

Measured

39.96 m/s reported for a car doing 20. The distance and the ordinary average were both correct at the same moment, which is why this could sit next to a readout that looks perfectly right.

Cause

The integrator decided whether the car was moving by asking whether that particular sample banked distance. Those are different questions. The noise gate deliberately holds small movements until they accumulate past the fix's own accuracy, so at 5 Hz one interval in two banks nothing while the car is moving perfectly steadily. Each of those was counted as a stop.

Worth noting

It got worse the faster the receiver, so a better GPS chip made the number less accurate. That is backwards, and it is the second defect in this codebase with that shape.

HIGH A revoked permission looked exactly like a tunnel

Before

The retry loop caught every platform error, which is correct, and then emitted the identical value a tunnel produces. So the failure became invisible the moment it was handled. A revoked permission, a dead sensor and a platform exception all read as `GPS_LOST`.

Why it matters

Those need opposite responses from the crew. A tunnel is waited out. A revoked permission has to be acted on, and the app was sending them to look for open sky.

After

The failure is carried on the sample and the cluster reads `GPS_ERROR` with its own icon. Retry behaviour is completely unchanged.

HIGH A ten minute reconnect fired inside a long tunnel

Before

A backstop tore down and rebuilt the location subscription after ten minutes of silence. The comment on that limit claimed it was longer than any real tunnel transit while, in the same sentence, citing the Lærdal tunnel at about 1,102 seconds against a 600 second limit. So it was guaranteed to fire in the longest road tunnel in the world, restarting the very foreground service that keeps the receiver alive in there.

After

Thirty minutes, which clears Lærdal at 60 km/h rather than only at its posted limit, because a tunnel is driven at the speed traffic allows. A reconnect triggered by silence alone is also no longer recorded as a fault, since no duration can prove a subscription is dead when a traffic jam can outlast any of them.

MINE A health counter that could never move

`STREAM_STALLS` had no caller at all, so it read zero forever. The road test procedure asks a tester to check that exact field, which meant the check could not fail and would have been signed off without measuring anything. This defect was introduced by me, not by you, and it is recorded here for the same reason as everything else.

Also fixed

- The true north calibration never actually learned, and displayed `MAG` while doing so, which reads as its own opposite.
- Agreeing samples were never checked for agreement. Twenty contradictory readings cleared the gate as readily as twenty consistent ones.

- The speedometer displayed readings the odometer had already rejected, so the two disagreed by construction.
- The GNSS health panel was frozen at whatever it read when the screen opened.
- The calibration relearned from zero on every launch.
- The GPS heading lagged by samples. Measured at 60.5 degrees of error at 5 Hz against 18.0 at 1 Hz for the same turn on the same road.
- `SafeArea` was missing on the map, which is the screen that runs landscape on a windscreen mount where the cutout sits on a side edge, exactly where the record button and coordinate readout are.
- Every settings row was dead to the touch. A coloured box between the row and its material ancestor swallowed the ink splash, so a tap gave no feedback at all.
- The trip readout physically shrank as you drove. Measured on a 360 by 110 tile, `9.99` renders 67.0 px tall and `100.00` renders 50.4 px, so crossing 100 km shrinks the number a co driver reads aloud by 25 percent and moves the decimal point 83 px. On a wide screen the same values differ by 0.1 px, which is why it was never caught by eye.
- Night mode never reached four screens, including the map coordinate bar and the split times, both of which sit in the driver's eyeline.
- The countdown was stuck at 1:00 because its setter was never called.

4. What the new tests can and cannot prove

Test count on its own is a weak number, so here is what the new coverage actually buys and where it stops.

Layer	Proves	Does not prove
Unit and widget 414 tests	Every measurement rule, the tunnel state machine, the compass source switch, and each defect above by a test that failed before its fix.	Anything about the real device, the real receiver, or real driving.
Golden images 4 images	Layout, colour and composition. They caught a night mode defect on their very first run.	Typography. The test framework substitutes a font whose glyphs are all one width, so they cannot see a wrong label or a wrong number.
Integration 5 tests, real device	The real boot path with real storage, real routing and real platform channels. It found the dead settings rows on its first complete run.	Anything needing GNSS. An emulator has no receiver, and injected location is synthetic.
iOS	It builds, installs, launches and lays out with no crash and no overflow. It found the permission screen telling iPhone users that Android needs a permission iOS never asks for.	Accuracy or compass. A simulator has neither GNSS nor a magnetometer.

No road test has ever happened, and this is the honest limit of everything above. Every accuracy figure in the repository comes from generated fixtures or an emulator, where the ground truth is exact by construction. `docs/ROAD-TEST.md` is the procedure. Its first item is a stop ship check: whether the trip counter can fail to come back after location services are switched off and on again. If that item fails, it reopens work rather than becoming a later fix.

5. Things deliberately left alone

- **The wakelock.** Reviewed and unchanged. Scoping it to a trip requires a concept of an active trip that the app does not have, and a cluster that blanks at a control is worse than a lit screen in a service park. The accepted cost is written into the code rather than only into a commit message.
- **Persian-Indic numerals.** For an Iranian crew the digits are the product. That is a market decision, not a code decision, so it is question 4 below.
- **Touch target on the speed unit toggle.** Enlarging a gesture nobody knows exists fixes the smaller half of the problem. Worth a design conversation before touching it.
- **Magnetic interference detection.** Investigated and reported as not built. The sensor package exposes no accuracy channel, so it is a native addition rather than something that can be read.

6. What I need from you

Two of these expire, in the sense that they get more expensive the longer they wait. The rest can be answered at your convenience.

Time sensitive

1. Whose signing key, and where is it kept?

The release APK is currently debug signed. An app first installed under a debug key **cannot be updated with a real one without uninstalling first**, and uninstalling wipes every tester's saved trips. This has to be settled before your father or anyone else builds up data worth keeping. Debug signing is also fatal for Cafe Bazaar, Myket and Play.

2. iOS 12 or iOS 13?

The Flutter iOS migrator raised `IPHONEOS_DEPLOYMENT_TARGET` from 12.0 to 13.0 on the first build. It is not optional, because current Flutter does not support iOS 12, but it changes which handsets can install the app and you had asked for a 12 floor. Confirm you accept 13, or we pin an older Flutter and accept that cost instead.

Product decisions

3. Which map tile source?

The offline maps TODO in the code cannot be completed as written. The OpenStreetMap Foundation policy explicitly prohibits the pre seeding it plans, so this is a licensing constraint before it is a preference. Options are ranked in `docs/IRAN-CONSTRAINTS.md`.

4. Persian-Indic digits, or Latin numerals with translated labels?

This is the one question I genuinely cannot answer for you. Ask a crew.

5. Is following an imported GPX route in scope?

Today a route can be imported, and the app confirms it parsed, and then it can only be exported again or deleted. **Nothing anywhere draws a saved route.** Importing a route strongly implies driving it, so as things stand this will get reported as a bug rather than a missing feature.

6. Should the trip counters be settable to a value?

They can be zeroed and nudged, never set. A regularity restart normally means entering a known distance.

7. Is portrait supported, or is the app landscape only?

It now launches portrait and follows the device rotation setting. Landscape is the clean layout and is what a mount uses.

Two readings of the specification, section 12.2

8. Does 12.2 saturate at the ceiling, or snap back to the entry speed?

The literal reading produces a sawtooth of 25, then 20, then 25, which on a driver facing readout looks like a fault rather than a limit.

9. Should it apply upwards only?

Applied symmetrically it would hold a stopped car at the entry speed and invent distance. The current behaviour lets braking run to a clean stop, and a test already asserts that.

7. What happens next

Item	Status
Two external code review passes	Run. All four findings on the measurement path are fixed.
Nine lower priority review findings	Open. Next block of work, none of them urgent.
Road test	Never done. Needs a car. Item 1 is stop ship.
Pull request	Not opened, on purpose. Yours and Saam's to open and merge.

Full technical detail with file references is in `docs/QA-REPORT.md` . The road test procedure is `docs/ROAD-TEST.md` . Every question above is also tracked in `docs/FOR-AMIRALI.md` with what would reverse each decision.